

The Quad- C_5 Graph: Maximum Contextuality Gap on Eight Vertices

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Quantum contextuality rules out classical descriptions in which measurement outcomes are independent of the compatible measurements performed alongside them. While the KCBS pentagon provides the minimal qutrit test, identifying larger but still compact contextuality witnesses remains important for understanding how contextuality can be organized, amplified, and benchmarked. We solve the eight-event version of this problem within the graph-theoretic framework for contextuality. An exhaustive search over all 11,117 connected non-isomorphic graphs with eight vertices identifies a sparse ten-edge graph, which we call Quad- C_5 , as the maximum-gap witness. The graph has a larger absolute contextuality gap than the Wagner graph, a standard eight-vertex benchmark, despite having fewer exclusivity relations. Its structure is transparent: it is formed from four overlapping KCBS pentagons, with each edge shared by two pentagons. We prove that Quad- C_5 is already contextual for a single qutrit by giving an explicit three-dimensional construction. In this qutrit realization its contextuality margin equals that of KCBS, whereas its full quantum advantage is reached numerically in four dimensions and exceeds that of the Wagner graph. Quad- C_5 therefore provides a compact bridge between minimal qutrit contextuality and stronger four-dimensional contextuality witnesses.

I. INTRODUCTION

Quantum contextuality [1] captures an irreducible tension between quantum mechanics and classical realism: measurement outcomes cannot, in general, be assigned pre-existing values independently of which compatible measurements are performed alongside them. Unlike Bell nonlocality, contextuality does not require spatially separated subsystems and can already appear in a single finite-dimensional quantum system. This makes it both a foundational signature of nonclassicality and a resource for quantum information processing, with established connections to quantum computation, communication, and sensing [2–4].

Identifying simple yet strong contextuality witnesses is a central goal in this program. The Klyachko–Can–Binicioğlu–Shumovsky (KCBS) construction demonstrated that five cyclically compatible measurements on a single spin-1 system are sufficient for a state-dependent contextuality test [5, 6]. This pentagonal structure became the canonical minimal qutrit example and was subsequently embedded in the broader family of n -cycle contextuality scenarios, where noncontextual inequalities and their quantum violations admit analytic characterization [7]. Such closed-form results are rare: separating classical, quantum, and more general probabilistic correlations is, in general, a hard problem.

Most experimental contextuality tests have focused on

minimal or highly structured scenarios, such as the KCBS pentagon for qutrits and n -cycle inequalities [7]. These examples are essential as foundational tests, but recent resource-theoretic developments motivate a broader question: how can contextuality be organized, amplified, and benchmarked in larger but still experimentally manageable exclusivity structures? This question is especially relevant because contextuality has been connected to advantages in quantum computation [3, 4], communication [8], and metrology [9]. From this viewpoint, small-graph optimization plays a role analogous to the search for compact Bell inequalities or small quantum error-correcting codes: one seeks the strongest nonclassical signature under limited resources. Here we address the eight-vertex absolute-gap optimization problem beyond the analytically understood cycle family and the standard Wagner benchmark.

A systematic approach to this optimization is provided by the graph-theoretic framework of Cabello, Severini, and Winter [10]. Each measurement event is represented as a vertex, and mutually exclusive events are connected by edges, yielding an exclusivity graph $G = (V, E)$. For projective measurements, the total event probability satisfies

$$\alpha(G) \leq \sum_{i \in V} p_i \leq \vartheta(G), \quad (1)$$

where $\alpha(G)$ is the independence number, giving the non-contextual bound, and $\vartheta(G)$ is the Lovász theta number [11], giving the projective quantum bound [10]. The contextuality gap

$$\Delta(G) = \vartheta(G) - \alpha(G) \quad (2)$$

measures the separation between quantum and noncontextual descriptions within this graph model; a larger gap

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provides a stronger contextuality resource [2].

Maximizing the contextuality gap for a fixed vertex count is a nontrivial combinatorial problem. Adding edges introduces additional exclusivity constraints that can suppress the quantum value, yet may also lower the classical bound and reorganize the geometry of the witness. Consequently, the densest or most symmetric graph need not yield the strongest contextuality, and even the eight-vertex case presents a non-obvious optimization landscape.

Known examples chart the structure of this hierarchy. The pentagon C_5 realizes the KCBS witness, and the seven-cycle C_7 belongs to the analytically characterized n -cycle family [7]. At eight vertices, the Wagner graph has served as a standard non-cycle benchmark [10, 11]. The full eight-vertex landscape, however, extends well beyond these symmetric examples, and it is not a priori clear whether any of these known graphs achieves the global maximum gap or whether a different, less symmetric structure outperforms them all.

A closely related graph-enumeration program was used in the study of fully contextual correlations, where the independence number, Lovász theta number, and fractional packing number were computed for nonisomorphic graphs with fewer than 11 vertices [12]. This database was later used to identify experimentally accessible graphs maximizing the ratio $\vartheta(G)/\alpha(G)$, a relative measure of contextuality strength [13]. The ratio $\vartheta(G)/\alpha(G)$ has since been studied as a graph-based dimension witness [31], and the CSW exclusivity-graph framework has been extended to linear games and Bell inequalities [30]. Building on this graph-theoretic viewpoint, we focus on a complementary optimization criterion: the absolute CSW gap $\Delta(G) = \vartheta(G) - \alpha(G)$. This criterion selects compact witnesses with the largest absolute quantum advantage, determined by the largest additive separation between the quantum and noncontextual bounds. We use it to identify the optimal connected eight-vertex graph and then analyze the structure, dimension dependence, and noise robustness of the resulting witness. Maximizing $\vartheta(G)/\alpha(G)$ favors large relative violations, often with smaller classical bounds, whereas maximizing $\vartheta(G) - \alpha(G)$ favors large additive separations between the quantum and noncontextual limits. These two objectives need not select the same graph, and at eight vertices they do not.

We resolve this question by an exhaustive semidefinite-programming search over all connected eight-vertex graphs. The numerically certified global maximum is attained by a ten-edge graph, which we call Quad- C_5 , that outperforms both the Wagner graph and every cycle in this vertex range. Its structure is transparent: four overlapping KCBS pentagons share edges in a regular pattern, providing a compact mechanism by which KCBS-type contextuality is efficiently combined within a larger exclusivity graph.

We additionally determine the minimal Hilbert-space dimension needed to realize this advantage. An explicit

three-dimensional orthogonal representation establishes that Quad- C_5 is already qutrit-contextual, and numerical evidence shows that its full graph-theoretic quantum value is first reached in four dimensions. The graph thus bridges minimal qutrit contextuality and the regime of stronger four-dimensional witnesses; we compare its noise robustness against both KCBS and Wagner.

The paper is organized as follows. Section II reviews the graph-theoretic background and the dimension-restricted quantum quantities used in the analysis. Section III describes the exhaustive graph search and numerical certification. Section IV presents the Quad- C_5 graph and its structural properties. Section V analyzes dimension dependence, noise robustness, and comparisons with KCBS and Wagner. Section VI discusses experimental implementation strategies and platform requirements. Section VII summarizes the conclusions. Appendix A provides the explicit qutrit construction establishing the analytic lower bound. Appendix B collects supplementary tables including the full edge lists, orthogonal representations, and numerical certification data.

II. MATHEMATICAL BACKGROUND

A. Lovász Theta and Semidefinite Programming

For a graph G on n vertices with edge set E , the Lovász theta function is given by the semidefinite program [11, 14]

$$\vartheta(G) = \max_{\substack{X \succeq 0, \text{Tr}(X)=1 \\ X_{ij}=0 \ \forall (i,j) \in E}} \mathbf{1}^\top X \mathbf{1}, \quad (3)$$

over the cone of $n \times n$ positive semidefinite matrices. Equivalently, via the orthonormal representation [11],

$$\vartheta(G) = \max_{|\psi\rangle, \{|v_i\rangle\}} \sum_{i=1}^n |\langle \psi | v_i \rangle|^2, \quad (4)$$

where $\{|v_i\rangle\} \subset \mathbb{C}^d$ are unit vectors with $\langle v_i | v_j \rangle = 0$ for $(i, j) \in E$, and $|\psi\rangle$ is a handle state. The Lovász sandwich theorem [11] gives $\alpha(G) \leq \vartheta(G) \leq \chi(\bar{G})$. The relaxation over real vectors is tight: $\vartheta(G)$ equals the maximum in Eq. (4) even when restricted to $|v_i\rangle \in \mathbb{R}^d$.

B. Quantum Dimension Hierarchy

The maximum quantum advantage achievable in Hilbert-space dimension d with real orthonormal representations is

$$\eta_d(G) = \max_{\substack{\{|v_i\rangle\} \subset \mathbb{R}^d, \|v_i\|=1, \\ \langle v_i | v_j \rangle = 0 \ \forall (i,j) \in E}} \lambda_{\max} \left(\sum_{i=1}^n |v_i\rangle\langle v_i| \right), \quad (5)$$

with $\eta_d(G) \nearrow \vartheta(G)$ as $d \rightarrow n$. The minimum dimension d^* is

$$d^*(G) = \min\{d : \eta_d(G) > \alpha(G)\}. \quad (6)$$

Throughout this paper we restrict to rank-one projectors as in the standard CSW formulation; contextuality witnesses using projectors of nonunit rank have been studied in Ref. [32].

We distinguish two witness types:

- **Qutrit witness** (here: $d^* = 3$, i.e., a graph for which $\eta_3(G) > \alpha(G)$): A single three-level system suffices. Note this usage is broader than “state-independent contextuality for spin-1 systems”; we mean only that $d^* = 3$.
- **(Numerically indicated) pure two-qubit witness**: $\eta_3(G) = \alpha(G)$ under exhaustive numerical search, and $\eta_4(G) > \alpha(G)$, indicating $d^* = 4$. Because η_d optimization is non-convex, the upper bound $\eta_3 \leq \alpha$ for specific graphs carries the caveat discussed in the Dimension Classification subsection.

III. METHODS

A. Exhaustive Graph Enumeration

McKay’s database [15] (graph8.g6) contains all 12,346 non-isomorphic simple graphs on $n = 8$ vertices; of these, 11,117 are connected (OEIS A001349). We retrieved all graphs, filtered for connectivity, and applied the analysis pipeline to the connected subset. Non-isomorphic graphs on $n = 5$ and $n = 7$ were obtained from NetworkX’s Graph Atlas [16].

Although related graph databases exist in the literature [12, 13, 15], we performed an independent enumeration and SDP calculation in order to optimize $\Delta(G) = \vartheta(G) - \alpha(G)$, verify the ranking, and obtain primal–dual numerical certificates for the leading graphs.

B. Independence Number

For each graph G we compute $\alpha(G) = \omega(\bar{G})$ via maximum-weight clique search on the complement, implemented in NetworkX. For $n \leq 8$ this is exact; we cross-validated all 11,117 results by independent brute-force enumeration of all $2^8 = 256$ vertex subsets, finding perfect agreement across every graph.

C. Lovász Theta via SDP

We solve Eq. (3) using CVXPY [17] with the SCS solver [18] for bulk screening of all 11,117 connected graphs; SCS provides typical precision of 10^{-4} – 10^{-6} , sufficient for ranking. The top 50 candidates were resolved with the CLARABEL solver (precision $\sim 10^{-10}$ – 10^{-12}), whose values are reported in all tables. For every graph in the top 50, the SCS–CLARABEL discrepancy

is $|\vartheta_{\text{SCS}} - \vartheta_{\text{CLARABEL}}| < 10^{-7}$ and the CLARABEL primal feasibility residual satisfies $r_p < 4 \times 10^{-10}$, providing numerical evidence against artefacts. The bulk scan completed in under four minutes on a standard laptop CPU.

To test the robustness of the ranking, we recomputed the leading candidates with two independent conic solvers and repeated the high-accuracy calculations five times. The resulting numerical ranges for the top three graphs are well separated: the lower endpoint for Quad- C_5 exceeds the upper endpoint for the second-ranked graph by 2.94×10^{-2} (Table I; see also Table V in Appendix B), many orders of magnitude larger than the reported primal and dual residuals. We therefore regard the ordering of the leading graphs, and in particular the uniqueness of Quad- C_5 as the eight-vertex maximizer, as numerically certified at the reported precision. A fully formal certificate would require exact arithmetic or independently verifiable interval primal–dual SDP certificates.

D. Dimension Classification via Constrained Optimization

Computing $\eta_d(G)$ is a non-convex optimization problem that yields a *lower bound* on η_d . We use a three-phase strategy with $N = 300$ random restarts:

1. *Phase 1* (L-BFGS-B): minimize orthogonality violations $\mathcal{L}_1 = \sum_{(i,j) \in E} \langle \hat{v}_i | \hat{v}_j \rangle^2$.
2. *Phase 2* (L-BFGS-B, penalty method): maximize $\lambda_{\max}(\sum_i |v_i\rangle\langle v_i|)$ with penalty weights $w_{\perp} = 50,000$, $w_{\lambda} = 0.3$, $w_{\text{norm}} = 5,000$.
3. *Phase 3* (SLSQP): refine with exact equality constraints.

The best $\lambda_{\max}$ over all restarts is a lower bound on η_d . For graphs where all 300 restarts converge to $\lambda_{\max} = \alpha$ (the classical bound), we report $\eta_d \leq \alpha$ as *numerically indicated*; a formal upper-bound certificate would require an SDP relaxation over d -dimensional representations, which is left for future work.

E. Closed-Form Identification via PSLQ

To identify algebraic closed forms, we apply the PSLQ algorithm [19] at 50-decimal-digit precision (mpmath library). We validate each candidate polynomial P of degree k via the residual test: the ratio

$$r = |P(x)| / (|P'(x)| \cdot \varepsilon_{\text{SDP}}) \quad (7)$$

must satisfy $r \ll 1$ for the relation to be genuine rather than a numerical coincidence ($\varepsilon_{\text{SDP}} \approx 10^{-15}$ is the CLARABEL precision). We search for integer relations of degree $k \leq 4$ with coefficient bound $|c_i| \leq 10^6$; the

TABLE I. Primal-dual certification for the top-3 connected $n = 8$ graphs (CLARABEL solver). ϑ_{CL} : primal objective value; $r_p = |\text{Tr } X - 1|$: trace-constraint residual; $r_{\text{edge}} = \max_{(i,j) \in E} |X_{ij}|$: maximum edge-constraint violation; $\lambda_{\min}(X)$: smallest eigenvalue of the primal matrix; $r_d = y - \vartheta_{\text{CL}}$: primal-dual objective gap; $\lambda_{\min}(S)$: smallest eigenvalue of the dual slack matrix. Full top-20 data in Appendix B.

Rank	Note	ϑ_{CL}	r_p	r_{edge}	$\lambda_{\min}(X)$	r_d	$\lambda_{\min}(S)$
1	Quad- C_5	3.46784	$+3.7 \times 10^{-17}$	$+3.7 \times 10^{-17}$	-5.6×10^{-9}	-8.5×10^{-9}	-2.5×10^{-8}
2		3.43845	$+1.2 \times 10^{-19}$	$+1.2 \times 10^{-19}$	-1.3×10^{-9}	-1.2×10^{-9}	-6.8×10^{-9}
3	Wagner W	3.41421	$+2.1 \times 10^{-19}$	$+2.1 \times 10^{-19}$	-5.7×10^{-9}	$+2.5 \times 10^{-9}$	-1.2×10^{-8}

unique solution at $k = 2$ is $x^2 - 2x - 4 = 0$. PSLQ detection is not a formal algebraic proof but constitutes high-confidence numerical identification [19]; a formal proof would require an independent algebraic construction or exact-arithmetic SDP.

IV. RESULTS

A. Reference Results: $n = 5$ and $n = 7$

For $n = 5$, the maximum contextuality gap is achieved by the five-cycle C_5 (the KCBS exclusion graph [6]):

$$\alpha(C_5) = 2, \quad \vartheta(C_5) = \sqrt{5}, \quad \Delta(C_5) = \sqrt{5} - 2 \approx 0.236. \quad (8)$$

For $n = 7$, the maximum is the seven-cycle C_7 [7], which we verify by exhaustive SDP search over all 853 connected non-isomorphic $n = 7$ graphs using the same pipeline:

$$\vartheta(C_7) = \frac{7 \cos(\pi/7)}{1 + \cos(\pi/7)} \approx 3.318, \quad \Delta(C_7) \approx 0.318. \quad (9)$$

Both C_5 and C_7 are qutrit witnesses ($d^* = 3$).

B. Exhaustive Search over $n = 8$

Table II lists the top-10 graphs by Δ from the exhaustive search. Quad- C_5 achieves $\Delta = 0.46784$, exceeding the Wagner graph W (Rank 3) by $0.46784 - 0.41421 = 0.05363$, while using only 10 edges compared to W 's 12.

C. Structure of the Quad- C_5 Graph

The graph6 canonical encoding [15] of this graph is GQQB'0, which we retain as a reproducibility identifier. Quad- C_5 has vertex set $V = \{0, 1, \dots, 7\}$ and edge set

$$E = \{(0, 3), (0, 5), (1, 4), (1, 6), (2, 5), (2, 6), (2, 7), (3, 6), (3, 7), (4, 7)\}. \quad (10)$$

Its degree sequence is $(2, 2, 2, 2, 3, 3, 3, 3)$: vertices $\{0, 1, 4, 5\}$ have degree 2 (“leaf” nodes) and vertices $\{2, 3, 6, 7\}$ have degree 3 (“hub” nodes). The graph is

TABLE II. Top-10 connected graphs on $n = 8$ vertices by contextuality gap $\Delta = \vartheta - \alpha$ (CLARABEL values), from exhaustive search over the 11,117 connected non-isomorphic simple 8-vertex graphs. The column ϑ/α is the relative contextuality measure used in earlier graph searches [13, 31]; its $n=8$ maximizer (Rank 7, †, the CHSH-type graph discussed in Ref. [13]) differs from the gap maximizer (Quad- C_5 , Rank 1).

Rank	Note	$ E $	α	ϑ	Δ	ϑ/α
1	Quad- C_5 ($n=8$ gap max)	10	3	3.46784	0.46784	1.15595
2		11	3	3.43845	0.43845	1.14615
3	Wagner W	12	3	3.41421	0.41421	1.13807
4		11	3	3.37228	0.37228	1.12409
5		12	3	3.37228	0.37228	1.12409
6		10	3	3.37228	0.37228	1.12409
7	† CHSH-type graph	16	2	2.34315	0.34315	1.17157
8		11	3	3.33804	0.33804	1.11268
9		17	2	2.33804	0.33804	1.16902
10		18	2	2.33333	0.33333	1.16667

neither regular nor vertex-transitive. The four hub nodes $\{2, 3, 6, 7\}$ form a complete bipartite subgraph $K_{2,2}$ (bipartition $\{2, 3\} \times \{6, 7\}$), while the leaf nodes attach as two pendant paths $3-0-5-2$ and $6-1-4-7$, each bridging one hub pair through two degree-2 vertices.

Figure 1 visualizes Quad- C_5 alongside the Wagner graph W for comparison.

A defining structural feature is that Quad- C_5 contains exactly four induced five-cycles (KCBS pentagons), listed in Table VI. Crucially, *every* one of the 10 edges belongs to exactly two of the four C_5 subgraphs—a perfect two-fold edge coverage.

The adjacency spectrum of Quad- C_5 is displayed in Table VII in Appendix B. Six of the eight eigenvalues belong to the golden-ratio field $\mathbb{Q}(\sqrt{5})$: $\{1 + \varphi, \varphi^{-1}(\times 2), 1 - \varphi^{-1}, -\varphi(\times 2)\}$, where $\varphi = (1 + \sqrt{5})/2$. This is a direct spectral fingerprint of the embedded C_5 subgraphs. The remaining two eigenvalues ($\approx +1.303$ and ≈ -2.303) do not lie in $\mathbb{Q}(\sqrt{5})$; their structural origin within the graph is an open question.

D. Edge Lists for the Top-6 Graphs

Complete edge lists for reproducibility are given in Table XII in Appendix B.

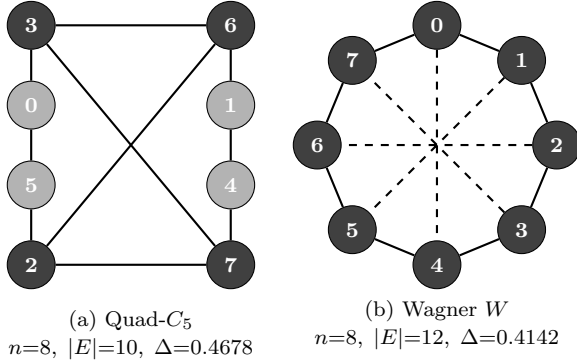


FIG. 1. (a) The optimal witness Quad- C_5 ($\Delta = 0.4678$, 10 edges). Dark (hub) nodes have degree 3; light (leaf) nodes have degree 2. The four hub nodes $\{2, 3, 6, 7\}$ form a $K_{2,2}$ core; leaves attach via two pendant paths $3-0-5-2$ (left column) and $6-1-4-7$ (right column). (b) The Wagner graph W ($\Delta = 0.4142$, 12 edges, 3-regular); dashed lines are the four “long diagonals” of the C_8 cycle. Quad- C_5 surpasses W by 13% in contextuality gap with two fewer edges.

E. Dimension Classification

Table VIII in Appendix B presents η_3 and dimension classifications. Ranks 1, 2, 4, and 6 are confirmed qutrit witnesses: the 300-restart optimization finds configurations with $\lambda_{\max} > 3$, providing a constructive lower bound $\eta_3 > \alpha$. The Wagner graph (Rank 3) and Rank 5 return $\lambda_{\max} = 3.000$ across all 300 restarts, consistent with $\eta_3 = \alpha = 3$ (numerically indicated pure two-qubit witnesses).

We note that the claim $\eta_3(W) = \alpha = 3$ is a *lower-bound* result: the optimization confirms that the best 3D representation found attains $\lambda_{\max} = \alpha$, but without an SDP upper-bound verification one cannot rule out a 3D configuration with $\lambda_{\max} > 3$ that the heuristic failed to find. For the Wagner graph, the high regularity and well-studied symmetry ($\text{Aut}(W) \cong D_4 \times \mathbb{Z}_2$, order 16) make the numerical finding credible, and a formal $d^* = 4$ proof remains an open question.

The numerically indicated η_3 value $1 + \sqrt{5} \approx 3.236$ appears for Ranks 1, 2, and 4; its analytic lower bound is established in the next subsection.

F. Orthogonal Vector Representations

Table IX in Appendix B summarizes the achieved $\lambda_{\max}$ values for the key configurations. For Quad- C_5 at $d = 3$, all 300 restarts converge to $\lambda_{\max} = 3.236068$ with maximum orthogonality error $< 2.3 \times 10^{-14}$, confirming the qutrit advantage is robustly attained. At $d = 4$, the optimization converges to $\lambda_{\max} = 3.467844 \approx \vartheta$, with orthogonality error $< 3.3 \times 10^{-14}$, confirming that the full Lovász bound is numerically saturated in four dimensions.

G. Analytic qutrit lower bound and numerical evidence for tightness

The optimized three-dimensional value found in all numerical runs is

$$\eta_3^{\text{num}}(\text{Quad-}C_5) = 1 + \sqrt{5} \approx 3.23607, \quad (11)$$

identified via PSLQ applied to the numerically obtained $\eta_3 \approx 3.236068$ at 50-digit precision. The degree-2 polynomial $P(x) = x^2 - 2x - 4$ has positive root $1 + \sqrt{5}$, and the residual test yields $r = |P(\eta_3)| / (|P'(\eta_3)| \cdot \varepsilon_{\text{SDP}}) \approx 0.3 \ll 1$, confirming a genuine algebraic relation. We prove the analytic lower bound

$$\eta_3(\text{Quad-}C_5) \geq 1 + \sqrt{5} \quad (12)$$

by constructing an explicit orthogonal representation in $\mathbb{R}^3$ (Appendix A). Extensive high-precision optimization and PSLQ recognition indicate that this bound is tight, although a formal dimension-constrained upper bound remains open.

A representative set of unit vectors $\{|v_i\rangle\} \subset \mathbb{R}^3$ achieving $\lambda_{\max} = 1 + \sqrt{5}$ (orthogonality error $< 2.5 \times 10^{-14}$, obtained from 500 independent restarts using SLSQP with explicit orthogonality constraints, all converging to the same $\lambda_{\max}$ with $|\lambda_{\max} - (1 + \sqrt{5})| < 3 \times 10^{-13}$) is given in Table IV.

This numerically determined value connects Quad- C_5 to the KCBS pentagon: both $\vartheta(C_5) = \sqrt{5}$ and $\eta_3(\text{Quad-}C_5) = 1 + \sqrt{5}$ lie in $\mathbb{Q}(\sqrt{5})$. The same η_3 value also appears for Ranks 2 and 4 (Table VIII in Appendix B).

By contrast, the candidate polynomial for $\vartheta(\text{Quad-}C_5)$ at 15-digit precision has $r \approx 20 \gg 1$ and is a false positive. A reliable closed form for ϑ requires a high-precision solver such as SDPA-GMP [20] at ≥ 30 significant digits.

H. Known-Graph Hierarchy

$$\Delta(C_5) < \Delta(C_7) < \Delta(W) < \Delta(\text{Quad-}C_5), \quad (13)$$

with numerical values in Table X in Appendix B.

V. DISCUSSION

A. Efficiency of Contextuality Geometry

The main surprise is that Quad- C_5 achieves a larger contextuality gap than W with *two fewer edges*. Adding an edge to an exclusion graph imposes an additional orthogonality constraint in Eq. (3), generically reducing $\vartheta(G)$ while leaving $\alpha(G)$ unchanged or reducing it. Naively, fewer edges should correlate with a smaller gap.

The fact that the optimal graph is sparser than the previously leading example points to a more efficient contextuality geometry: the four interlocking KCBS pentagons in Quad- C_5 encode more quantum advantage per edge than the symmetric 3-regular structure of W .

The two-fold edge coverage by the four C_5 subgraphs (Table VI in Appendix B) is a candidate structural explanation, though a formal proof of this connection remains open. This is related to properties of the Lovász theta under graph combinations [11]: shared edges prevent full cancellation of the individual contributions, allowing the four C_5 copies to reinforce each other.

B. Physical Significance of the Qutrit–Two-Qubit Distinction

A qutrit witness ($d^* = 3$) is testable on a single three-level system: photon orbital angular momentum [21] or an atomic Λ -system [22]. Such systems avoid the engineering complexity of entangled bipartite platforms and are more resilient to certain decoherence channels.

A (numerically indicated) pure two-qubit witness ($d^* = 4$) cannot exploit contextuality in any three-dimensional Hilbert-space embedding: the orthogonality constraints for its edge set require a four-dimensional space. This makes such witnesses more demanding experimentally but potentially more useful as probes of composite-system structure. Contextuality of this kind remains distinct from Bell nonlocality, as no entanglement between the two qubits is required [23, 33].

Remarkably, W and Rank 5 are non-isomorphic graphs but share identical numerically indicated dimensional behavior ($\eta_3 = \alpha$, $\eta_4 > \alpha$). This structural coincidence suggests an underlying algebraic invariant beyond ϑ and α that governs dimensional rigidity.

C. Noise Robustness under Depolarizing Noise

We analyze noise robustness via the standard depolarizing channel $\rho_{\text{noisy}} = v \rho + (1-v) I/d$, where $v \in [0, 1]$ is the visibility. Since each observable is a rank-1 projector with $\text{Tr}(P_i) = 1$, the observable sum under this channel becomes $S(v) = v \eta_d(G) + (1-v) n/d$, and contextuality is certified whenever $S(v) > \alpha(G)$. Solving for the minimum visibility required yields the *critical visibility*

$$v^* = \frac{\alpha(G) - n/d}{\eta_d(G) - n/d}, \quad (14)$$

with noise tolerance $1 - v^*$. Table III lists the values for the witnesses studied here.

The most striking entry in Table III is the exact coincidence $v^*(\text{KCBS}) = v^*(\text{Quad-}C_5, d=3)$: despite having three more vertices, Quad- C_5 at $d = 3$ tolerates up to 41.5% depolarizing noise—the same threshold as the simplest known contextuality witness. Substituting the

TABLE III. Critical visibility v^* and noise tolerance $1 - v^*$ under depolarizing noise, computed from Eq. (14). The unrestricted Lovász values ϑ are obtained from the high-accuracy SDP calculations in Table V in Appendix B. The dimension-constrained η_3 values are obtained from the non-convex orthogonal-representation optimization described in Sec. III D.

Graph	n	α	η_d	d	v^*	$1 - v^*$
C_5 (KCBS)	5	2	2.23607	3	0.5854	0.4146
C_7	7	3	3.31767	3	0.6773	0.3227
Quad- C_5 ($d = 3$)	8	3	3.23607	3	0.5854	0.4146
Quad- C_5 ($d = 4$)	8	3	3.46784	4	0.6813	0.3187
Wagner W ($d = 4$)	8	3	3.41421	4	0.7071	0.2929

established values into Eq. (14),

$$v^*(\text{KCBS}) = \frac{2 - \frac{5}{3}}{\sqrt{5} - \frac{5}{3}} = \frac{1}{3\sqrt{5} - 5}, \quad (15)$$

$$v^*(\text{Quad-}C_5, d=3) = \frac{3 - \frac{8}{3}}{(1 + \sqrt{5}) - \frac{8}{3}} = \frac{1}{3\sqrt{5} - 5}, \quad (16)$$

confirming the equality analytically. The coincidence is explained by a *uniform shift*: going from C_5 to Quad- C_5 , all three parameters entering Eq. (14) increase by the same amount, $\Delta\alpha = \Delta\eta_3 = \Delta(n/d) = 1$. Since the ratio in Eq. (14) is invariant under any common additive shift k ,

$$\frac{(\alpha + k) - (n/d + k)}{(\eta + k) - (n/d + k)} = \frac{\alpha - n/d}{\eta - n/d}, \quad (17)$$

the critical visibilities are identical. The shift $\Delta(n/d) = 1$ follows immediately: Quad- C_5 has $n = 8 = 5 + d$ vertices. Why $\Delta\alpha = \Delta\eta_3 = 1$ hold simultaneously is an open structural question not resolved by the exhaustive search.

This equality has a direct experimental consequence: any qutrit apparatus sufficient for a KCBS test can certify Quad- C_5 contextuality without any relaxation of the noise requirements. At $d = 3$ both witnesses achieve the same quantum advantage $\eta_3 - \alpha = \sqrt{5} - 2 \approx 0.236$; the larger Lovász gap $\Delta(\text{Quad-}C_5) = 0.468$ reflects the full quantum potential of the graph and is accessible at $d = 4$.

At $d = 4$, Quad- C_5 also strictly outperforms Wagner in noise robustness. The numerically indicated value $\vartheta(W) = 2 + \sqrt{2}$ yields $v^*(W, d=4) = 1/\sqrt{2} \approx 0.707$, whereas $v^*(\text{Quad-}C_5, d=4) \approx 0.681$, a reduction of 2.6%. Any experiment achieving visibility $v \in [0.681, 0.707]$ can therefore certify Quad- C_5 contextuality but not Wagner contextuality.

Finally, the C_7 entry in Table III illustrates that a larger gap Δ does not guarantee better noise robustness. Despite $\Delta(C_7) = 0.318 > \Delta(\text{KCBS}) = 0.236$, C_7 requires strictly higher visibility ($v^* = 0.677 > 0.585$). The reason is structural: the larger vertex count raises the mixed-state baseline $n/d = 7/3$, compressing the gap between the classical bound and the quantum value and eroding the advantage faster under noise. The critical

visibility v^* , not Δ alone, is the operationally correct figure of merit for experimental feasibility.

D. Algebraic Connection to KCBS via $\mathbb{Q}(\sqrt{5})$

The high-confidence identification $\eta_3(\text{Quad-}C_5) = 1 + \sqrt{5}$ reveals a deep algebraic link to the KCBS scenario. Both $\vartheta(C_5) = \sqrt{5}$ and $\eta_3(\text{Quad-}C_5) = 1 + \sqrt{5}$ lie in $\mathbb{Q}(\sqrt{5})$. Equivalently, the three-dimensional advantage of $\text{Quad-}C_5$ above the classical bound $\alpha = 3$ is exactly $\sqrt{5}$ —one unit of KCBS advantage above the classical floor.

The adjacency spectrum reinforces this picture: the six golden-ratio eigenvalues $\{1 + \varphi, \varphi^{-1}(\times 2), 1 - \varphi^{-1}, -\varphi(\times 2)\}$ are the spectral fingerprint of graphs built from C_5 building blocks [24], providing spectral-graph-theory evidence that the contextuality advantage is inherited from the KCBS subgraphs.

VI. EXPERIMENTAL FEASIBILITY

From a fundamental standpoint, KCBS is the simplest qutrit witness, requiring only five events; this also makes it attractive for experimental tests of foundational contextuality [6, 22]. On the other hand, contextuality can have practical value and hence benefit from scaling. The $\text{Quad-}C_5$ graph addresses how contextuality can be organized and enhanced in a compact eight-event exclusivity structure.

A qutrit implementation requires eight rank-one projectors

$$\Pi_i = |v_i\rangle\langle v_i|, \quad i = 0, \dots, 7, \quad (18)$$

satisfying

$$\Pi_i \Pi_j = 0 \quad \text{for every edge } (i, j) \in E. \quad (19)$$

The explicit construction in Appendix A provides such a representation and therefore gives a direct experimental prescription. The optimal input state is the eigenstate of

$$M = \sum_i \Pi_i \quad (20)$$

with largest eigenvalue. In the qutrit construction this state yields

$$\sum_i \langle \Pi_i \rangle = 1 + \sqrt{5} \quad (21)$$

(see Eq. (A4) in Appendix A).

The qutrit realization is non-injective: two pairs of nonadjacent vertices are represented by identical projectors ($v_0 = v_6$ and $v_2 = v_3$). Operationally, this corresponds to repeated projector terms appearing in different exclusivity contexts. This is consistent with the graph-theoretic orthogonal-representation framework, but it

also shows why the qutrit advantage remains KCBS-like. The larger contextuality gap of $\text{Quad-}C_5$ is obtained only in a four-dimensional realization, where the full Lovász value is reached numerically.

Possible platforms include spin-1 ion-trap and solid-state systems [6, 25, 26], photonic path or orbital-angular-momentum qutrits [21, 22, 27], NMR qutrits [2, 28], and four-level photonic or two-qubit systems [29]. In a four-dimensional implementation, $\text{Quad-}C_5$ becomes experimentally more appealing than the Wagner benchmark because it requires a strictly lower critical visibility under depolarizing noise ($v^* \approx 0.681$ vs. $v^*(W) \approx 0.707$).

VII. CONCLUSION

We have carried out an exhaustive search over all 11,117 connected non-isomorphic simple graphs on eight vertices and identified $\text{Quad-}C_5$ —an 8-vertex, 10-edge graph containing four interlocking KCBS pentagons—as the numerically certified unique maximizer of the contextuality gap for $n = 8$. With $\Delta = 0.46784$, it surpasses the Wagner graph ($\Delta \approx 0.414$) by $\sim 13\%$ while using two fewer edges.

Unlike the Wagner graph, which numerical evidence indicates requires a four-dimensional Hilbert space, $\text{Quad-}C_5$ is a confirmed qutrit witness: high-precision numerics indicate that $\eta_3(\text{Quad-}C_5) = 1 + \sqrt{5}$, placing this value in the same algebraic field $\mathbb{Q}(\sqrt{5})$ as the KCBS pentagon. In four dimensions, the full Lovász bound is saturated with orthogonality error below 10^{-13} .

A depolarizing noise analysis reveals that $\text{Quad-}C_5$ at $d = 3$ shares the critical visibility $v^* = 1/(3\sqrt{5} - 5) \approx 0.585$ of the KCBS witness, an equality that follows analytically from a uniform shift of the graph parameters α , η_3 , and n/d . Any qutrit apparatus sufficient for a KCBS test can therefore certify $\text{Quad-}C_5$ contextuality without relaxing noise requirements. At $d = 4$, $\text{Quad-}C_5$ also strictly outperforms the Wagner graph in noise robustness, requiring 2.6% less visibility.

The present optimization is complementary to earlier graph-based searches for fully contextual correlations [12] and for graphs maximizing the relative measure $\vartheta(G)/\alpha(G)$ [13]. Those searches identify different extremal structures. Here the objective is the absolute CSW gap $\Delta(G) = \vartheta(G) - \alpha(G)$, for which $\text{Quad-}C_5$ is the eight-vertex maximizer in our exhaustive search.

These results show that the optimal eight-vertex exclusivity graph is already qutrit-contextual, while its full Lovász value gives a stronger four-dimensional contextuality benchmark than the Wagner graph. They provide concrete graph structures suitable for experimental tests in three-level and four-level quantum systems. The finding that the $n = 8$ maximum requires *fewer* edges than the Wagner graph suggests that denser exclusion graphs do not always yield stronger contextuality; the relationship between graph sparsity and contextuality strength warrants systematic future investigation.

DATA AVAILABILITY

All graph data derive from the publicly available McKay database (graph8.g6, <https://users.cecs.anu.edu.au/~bdm/data/>). Complete edge lists are given in Table XII (Appendix B); All Python scripts and data files are publicly available at <https://github.com/ugurtamerphys/quad-c5-contextuality> and archived on Zenodo [34].

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Appendix A: Analytic Lower Bound:

$$\eta_3(\text{Quad-}C_5) \geq 1 + \sqrt{5}$$

We construct an explicit orthogonal representation of Quad- C_5 in $\mathbb{R}^3$ achieving $\lambda_{\max} = 1 + \sqrt{5}$, thereby proving $\eta_3(\text{Quad-}C_5) \geq 1 + \sqrt{5}$.

1. Graph Structure

The Quad- C_5 graph has vertex set $\{0, \dots, 7\}$ and edge set

$$E = \{(0, 3), (0, 5), (1, 4), (1, 6), (2, 5), (2, 6), (2, 7), (3, 6), (3, 7), (4, 7)\}.$$

Figure 2 shows the graph together with its four embedded C_5 pentagons.

It contains exactly four induced C_5 pentagons, shown in Fig. 3: $C_5^{(1)}$: 0-3-6-2-5-0; $C_5^{(2)}$: 0-3-7-2-5-0; $C_5^{(3)}$: 1-4-7-2-6-1; $C_5^{(4)}$: 1-4-7-3-6-1. Every edge of Quad- C_5 belongs to exactly two of these pentagons.

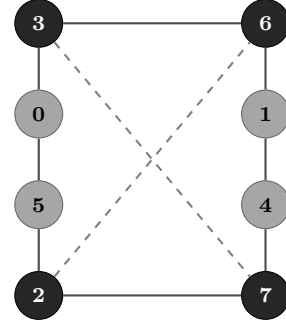


FIG. 2. The Quad- C_5 graph ($n = 8, m = 10$) in the same layout as Fig. 1(a). Dark (hub) nodes $\{2, 3, 6, 7\}$ have degree 3; light (leaf) nodes $\{0, 1, 4, 5\}$ have degree 2. Dashed edges are the two $K_{2,2}$ diagonals $(2, 6)$ and $(3, 7)$.

2. Explicit Construction

Let $\varphi = (1 + \sqrt{5})/2$ denote the golden ratio. Define the following unit vectors in $\mathbb{R}^3$:

$$\begin{aligned} v_0 &= v_6 = (1, 0, 0), \\ v_1 &= (0, \varphi^{-1/2}, \varphi^{-1}), \\ v_2 &= v_3 = (0, 0, 1), \\ v_4 &= (-\varphi^{-1}, -\varphi^{-3/2}, \varphi^{-1}), \\ v_5 &= (0, 1, 0), \\ v_7 &= (-\varphi^{-1}, \varphi^{-1/2}, 0), \end{aligned} \tag{A1}$$

where $\varphi^{-1/2} = \sqrt{(\sqrt{5} - 1)/2}$, $\varphi^{-1} = (\sqrt{5} - 1)/2$, and $\varphi^{-3/2} = \varphi^{-1} \cdot \varphi^{-1/2}$. Note $v_0 = v_6$ and $v_2 = v_3$; the pairs are coincident because $(0, 6) \notin E$ and $(2, 3) \notin E$.

All coordinates lie in $\mathbb{Q}(\sqrt{5}, \varphi^{-1/2})$.

Proposition A.1. *The vectors in Eq. (A1) satisfy: (i) $|v_i| = 1$ for all i ; and (ii) $\langle v_i | v_j \rangle = 0$ for every edge $(i, j) \in E$.*

Proof. Unit norms. Using $\varphi^{-1} + \varphi^{-2} = 1$ (golden ratio identity): $|v_1|^2 = \varphi^{-1} + \varphi^{-2} = 1$. For v_4 : $|v_4|^2 = \varphi^{-2} + \varphi^{-2} \cdot \varphi^{-1} + \varphi^{-2} = \varphi^{-2}(2 + \varphi^{-1}) = (2\varphi + 1)/\varphi^3 = \varphi^3/\varphi^3 = 1$, using $2\varphi + 1 = \varphi^3$. For v_7 : $|v_7|^2 = \varphi^{-2} + \varphi^{-1} = (3 - \sqrt{5})/2 + (\sqrt{5} - 1)/2 = 1$.

Orthogonality. All ten constraints are verified directly: $\langle v_0 | v_3 \rangle = \langle e_1 | e_3 \rangle = 0$; $\langle v_0 | v_5 \rangle = \langle e_1 | e_2 \rangle = 0$; $\langle v_1 | v_4 \rangle = -\varphi^{-2} + \varphi^{-2} = 0$; $\langle v_1 | v_6 \rangle = \langle v_1 | e_1 \rangle = 0$; $\langle v_2 | v_5 \rangle = \langle e_3 | e_2 \rangle = 0$; $\langle v_2 | v_6 \rangle = \langle e_3 | e_1 \rangle = 0$; $\langle v_2 | v_7 \rangle = 0$ (since v_7 has no z -component); $\langle v_3 | v_6 \rangle = \langle v_3 | v_7 \rangle = 0$ (same reasoning); $\langle v_4 | v_7 \rangle = \varphi^{-2} - \varphi^{-1} \cdot \varphi^{-1} = 0$. $\square$

3. Spectral Analysis

Let $M = \sum_{i=0}^7 |v_i\rangle\langle v_i|$. With $v_0 = v_6$ and $v_2 = v_3$, a direct computation gives $\text{Tr}(M) = 8$ and characteristic

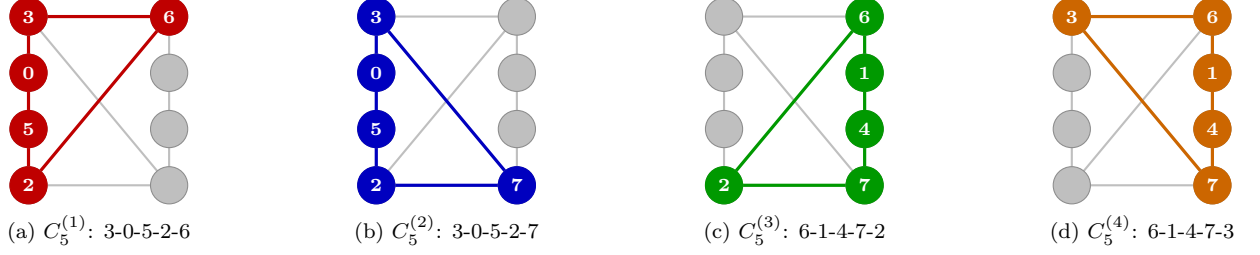


FIG. 3. The four induced C_5 pentagons in Quad- C_5 , using the same vertex layout as Fig. 1(a). Colored edges and vertices belong to each pentagon; every edge of Quad- C_5 appears in exactly two of the four pentagons.

TABLE IV. Representative unit vectors forming an orthogonal representation $\{|v_i\rangle\} \subset \mathbb{R}^3$ achieving $\eta_3(\text{Quad-}C_5) = 1 + \sqrt{5}$. Orthogonality error $\max_{(i,j) \in E} |\langle v_i | v_j \rangle| < 2.5 \times 10^{-14}$. All 500 independent restarts converged to $\lambda_{\max} = 3.236068$.

i	$v_i^{(1)}$	$v_i^{(2)}$	$v_i^{(3)}$
0	-0.327925	-0.827319	+0.456080
1	+0.591738	+0.502702	+0.630188
2	+0.265430	-0.544016	-0.795986
3	+0.265430	-0.544016	-0.795986
4	-0.676200	+0.735118	+0.048537
5	+0.906649	-0.139967	+0.397992
6	-0.072902	+0.811912	-0.579210
7	+0.710727	+0.668271	-0.219730

Note: $v_2 = v_3$ since $(2, 3) \notin E$; this is a valid non-injective representation.

polynomial

$$p(\lambda) = \lambda^3 - 8\lambda^2 + \left(\frac{31}{2} + \frac{5\sqrt{5}}{2}\right)\lambda + 4 - 10\sqrt{5}. \quad (\text{A2})$$

Substituting $\lambda = 1 + \sqrt{5}$ and using $(1 + \sqrt{5})^2 = 6 + 2\sqrt{5}$, $(1 + \sqrt{5})^3 = 16 + 8\sqrt{5}$:

$$\begin{aligned} p(1 + \sqrt{5}) &= (16 + 8\sqrt{5}) - (48 + 16\sqrt{5}) \\ &\quad + (28 + 18\sqrt{5}) + (4 - 10\sqrt{5}) = 0. \end{aligned}$$

Factoring out $(\lambda - (1 + \sqrt{5}))$ yields

$$p(\lambda) = (\lambda - (1 + \sqrt{5})) \left(\lambda - \frac{7 - \sqrt{5}}{2} \right)^2, \quad (\text{A3})$$

where the double root is $(7 - \sqrt{5})/2 \approx 2.382$. Since $(1 + \sqrt{5}) - (7 - \sqrt{5})/2 = (3\sqrt{5} - 5)/2 > 0$, the largest eigenvalue is $\lambda_{\max}(M) = 1 + \sqrt{5}$, proving $\eta_3(\text{Quad-}C_5) \geq 1 + \sqrt{5}$.

4. Critical State

The state attaining the maximum is the normalized eigenvector

$$|\psi_\star\rangle = \frac{1}{5^{1/4}} \begin{pmatrix} -1 \\ \sqrt{\sqrt{5} - 2} \\ 1 \end{pmatrix}, \quad (\text{A4})$$

which satisfies $M|\psi_\star\rangle = (1 + \sqrt{5})|\psi_\star\rangle$. Therefore

$$\sum_{i=0}^7 |\langle \psi_\star | v_i \rangle|^2 = \langle \psi_\star | M | \psi_\star \rangle = 1 + \sqrt{5}. \quad (\text{A5})$$

5. Note on Non-Injectivity

We emphasize that the qutrit construction in Appendix A is a valid orthogonal representation of the exclusivity graph, but it is not injective: two nonadjacent vertex pairs are represented by identical projectors, $v_0 = v_6$ and $v_2 = v_3$. Operationally, this corresponds to repeated projector terms appearing in different exclusivity contexts. This explains why the qutrit contextuality margin remains KCBS-like, $\eta_3 - \alpha = \sqrt{5} - 2$. The genuinely larger gap, $\vartheta - \alpha \simeq 0.46784$, is obtained only when the full Lovász value is accessed, which our numerical optimization realizes in four dimensions. Thus Quad- C_5 serves as an eight-event graph whose qutrit realization retains KCBS-type contextuality, while its full quantum realization exceeds the Wagner benchmark.

Appendix B: Supplementary Tables

TABLE V. Numerical ϑ ranges from 5×2 independent solver runs (SCS + CLARABEL). The gap between the Quad- C_5 lower endpoint and the Rank-2 upper endpoint is 2.94×10^{-2} , far larger than the observed solver residuals. This provides high-precision numerical evidence for uniqueness of the maximizer at the reported precision.

Graph	Lower bound	Upper bound	Half-width
Quad- C_5 (Rank 1)	3.46784373	3.46784378	2.4×10^{-8}
Rank 2	3.43844718	3.43844719	6.3×10^{-10}
Wagner W	3.41421356	3.41421364	3.7×10^{-8}

TABLE VI. The four induced C_5 subgraphs of Quad- C_5 . Every edge of the full graph belongs to exactly two pentagons.

Label	Vertices	Edges
$C_5^{(1)}$	$\{0, 2, 3, 5, 6\}$	$(0, 3), (0, 5), (2, 5), (2, 6), (3, 6)$
$C_5^{(2)}$	$\{0, 2, 3, 5, 7\}$	$(0, 3), (0, 5), (2, 5), (2, 7), (3, 7)$
$C_5^{(3)}$	$\{1, 2, 4, 6, 7\}$	$(1, 4), (1, 6), (2, 6), (2, 7), (4, 7)$
$C_5^{(4)}$	$\{1, 3, 4, 6, 7\}$	$(1, 4), (1, 6), (3, 6), (3, 7), (4, 7)$

TABLE VII. Adjacency spectrum of Quad- C_5 with $\varphi = (1 + \sqrt{5})/2$.

λ	Closed form	Mult.	In $\mathbb{Q}(\sqrt{5})$?
+2.618034	$1 + \varphi = \varphi^2$	1	Yes
+1.302776	—	1	No
+0.618034	φ^{-1}	2	Yes
+0.381966	$1 - \varphi^{-1}$	1	Yes
-1.618034	$-\varphi$	2	Yes
-2.302776	—	1	No

TABLE VIII. Dimension classification of the six leading $n = 8$ contextuality witnesses. η_3 is a lower bound on $\eta_3(G)$; d^* is numerically indicated (qutrit cases are confirmed; two-qubit cases are indicated by 300-restart convergence to $\lambda_{\max} = \alpha$).

Rank	Δ	η_3 (lb)	d^*	Type
1 (Quad- C_5)	0.46784	3.23607	3	Qutrit witness
2	0.43845	3.23607	3	Qutrit witness
3 (Wagner)	0.41421	3.00000	4 [†]	Indicated 2-qubit
4	0.37228	3.23607	3	Qutrit witness
5	0.37228	3.00000	4 [†]	Indicated 2-qubit
6	0.37228	3.33804	3	Qutrit witness

[†] Numerically indicated; upper bound not proved here.

TABLE IX. Orthogonal representation results. $\lambda_{\max}$ is a lower bound on $\eta_d(G)$; orth. error is the maximum $|\langle v_i | v_j \rangle|$ over edge pairs at convergence. “Adv.” means $\lambda_{\max} > \alpha = 3$.

Configuration	d	$\lambda_{\max}$	Adv.	Orth. error
Quad- C_5 , $d = 3$	3	3.236068	Yes	$< 2.3 \times 10^{-14}$
Quad- C_5 , $d = 4$	4	3.467844	Yes	$< 3.3 \times 10^{-14}$
Wagner, $d = 3$	3	3.000000	No	$< 6.0 \times 10^{-15}$
Wagner, $d = 4$	4	3.414214	Yes	$< 5.8 \times 10^{-14}$
Rank 5, $d = 3$	3	3.000000	No	$< 6.0 \times 10^{-15}$
Rank 5, $d = 4$	4	> 3	Yes	$< 10^{-13}$

TABLE X. Contextuality gap hierarchy for canonical small graphs. $d^* = 3$ is proved for C_5 and C_7 , and confirmed by $\eta_3 > \alpha$ for Quad- C_5 ; $d^* = 4$ for W is numerically indicated.

Graph	n	α	ϑ	Δ	d^*
C_5 (KCBS)	5	2	2.23607	0.23607	3
C_7	7	3	3.31767	0.31767	3
Wagner W	8	3	3.41421	0.41421	4 [†]
Quad-C_5	8	3	3.46784	0.46784	3

[†] Numerically indicated.

TABLE XI. Full primal-dual certification for the top-20 connected $n = 8$ graphs (CLARABEL solver). ϑ_{CL} : primal objective value; $r_p = |\text{Tr } X - 1|$: trace-constraint residual; $r_{\text{edge}} = \max_{(i,j) \in E} |X_{ij}|$: maximum edge-constraint violation; $\lambda_{\min}(X)$: smallest eigenvalue of the primal matrix; $r_d = y - \vartheta_{\text{CL}}$: primal-dual objective gap; $\lambda_{\min}(S)$: smallest eigenvalue of the dual slack matrix $S = yI + \sum_{(i,j) \in E} z_{ij} B_{ij} - J$. All solver statuses are **optimal**; inter-solver discrepancy $|\vartheta_{\text{SCS}} - \vartheta_{\text{CL}}| < 10^{-7}$ for all 50 graphs.

Rank	Note	ϑ_{CL}	r_p	r_{edge}	$\lambda_{\min}(X)$	r_d	$\lambda_{\min}(S)$
1	Quad- C_5	3.46784	$+3.7 \times 10^{-17}$	$+3.7 \times 10^{-17}$	-5.6×10^{-9}	-8.5×10^{-9}	-2.5×10^{-8}
2		3.43845	$+1.2 \times 10^{-19}$	$+1.2 \times 10^{-19}$	-1.3×10^{-9}	-1.2×10^{-9}	-6.8×10^{-9}
3	Wagner W	3.41421	$+2.1 \times 10^{-19}$	$+2.1 \times 10^{-19}$	-5.7×10^{-9}	$+2.5 \times 10^{-9}$	-1.2×10^{-8}
4		3.37228	$+4.4 \times 10^{-16}$	$+3.7 \times 10^{-17}$	-4.8×10^{-9}	-5.6×10^{-9}	$+4.9 \times 10^{-9}$
5		3.37228	$+4.2 \times 10^{-18}$	$+4.3 \times 10^{-18}$	-2.6×10^{-9}	-3.3×10^{-9}	$+2.0 \times 10^{-9}$
6		3.37228	$+8.8 \times 10^{-15}$	$+1.8 \times 10^{-15}$	-3.9×10^{-10}	-4.6×10^{-10}	-2.6×10^{-9}
7		2.34315	$+3.1 \times 10^{-15}$	$+9.4 \times 10^{-16}$	-6.6×10^{-11}	$+1.3 \times 10^{-11}$	-1.5×10^{-10}
8		3.33804	$+1.4 \times 10^{-14}$	$+1.4 \times 10^{-14}$	-6.6×10^{-10}	-1.6×10^{-9}	-3.2×10^{-9}
9		2.33804	$+2.2 \times 10^{-16}$	$+3.4 \times 10^{-18}$	-2.3×10^{-9}	$+3.0 \times 10^{-10}$	-3.8×10^{-9}
10		2.33333	$+5.6 \times 10^{-18}$	$+5.7 \times 10^{-18}$	-7.7×10^{-10}	-3.4×10^{-10}	-2.5×10^{-9}
11		3.33097	$+1.9 \times 10^{-19}$	$+1.9 \times 10^{-19}$	-1.1×10^{-9}	-2.0×10^{-9}	$+1.9 \times 10^{-9}$
12		3.32922	$+4.1 \times 10^{-18}$	$+4.1 \times 10^{-18}$	-4.3×10^{-9}	-7.5×10^{-9}	$+7.8 \times 10^{-9}$
13		3.31767	$+9.2 \times 10^{-19}$	$+9.2 \times 10^{-19}$	-2.2×10^{-9}	-5.2×10^{-9}	-1.7×10^{-8}
14		3.31767	$+9.4 \times 10^{-19}$	$+9.4 \times 10^{-19}$	-1.7×10^{-9}	-4.9×10^{-9}	-9.5×10^{-9}
15		3.31767	$+2.2 \times 10^{-16}$	$+2.7 \times 10^{-18}$	-1.5×10^{-9}	-2.1×10^{-9}	-5.6×10^{-9}
16		3.31767	$+4.9 \times 10^{-11}$	$+4.9 \times 10^{-11}$	-2.8×10^{-9}	-8.1×10^{-9}	-2.0×10^{-8}
17		3.31767	$+3.1 \times 10^{-10}$	$+3.1 \times 10^{-10}$	-1.9×10^{-9}	-3.4×10^{-9}	-6.5×10^{-9}
18		3.31767	$+3.1 \times 10^{-11}$	$+3.1 \times 10^{-11}$	-3.3×10^{-9}	-4.4×10^{-9}	-1.7×10^{-8}
19		3.31767	$+1.2 \times 10^{-13}$	$+2.3 \times 10^{-15}$	-8.1×10^{-10}	-2.3×10^{-9}	-2.2×10^{-9}
20		3.31767	$+4.1 \times 10^{-11}$	$+4.1 \times 10^{-11}$	-8.9×10^{-10}	-1.7×10^{-9}	-7.6×10^{-10}

TABLE XII. Edge lists for the six highest-ranked connected 8-vertex contextuality witnesses. All Δ values from CLARABEL.

Rank	Δ	d^*	$ E $	Edge list
1	0.46784	3	10	(0, 3), (0, 5), (1, 4), (1, 6), (2, 5), (2, 6), (2, 7), (3, 6), (3, 7), (4, 7)
2	0.43845	3	11	(0, 3), (0, 5), (1, 4), (1, 5), (1, 7), (2, 5), (2, 6), (2, 7), (3, 6), (3, 7), (4, 6)
3 (Wagner W)	0.41421	$4^\dagger$	12	(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7), (7, 0), (0, 4), (1, 5), (2, 6), (3, 7)
4	0.37228	3	11	(0, 3), (0, 5), (0, 7), (1, 4), (1, 6), (2, 5), (2, 6), (2, 7), (3, 6), (3, 7), (4, 7)
5	0.37228	$4^\dagger$	12	(0, 3), (0, 5), (1, 4), (1, 5), (1, 6), (2, 5), (2, 6), (2, 7), (3, 6), (3, 7), (4, 7), (5, 7)
6	0.37228	3	10	(0, 3), (0, 4), (0, 7), (1, 4), (1, 5), (2, 5), (2, 6), (2, 7), (3, 6), (4, 7)

† Numerically indicated; see the Dimension Classification subsection.

- [1] S. Kochen and E. P. Specker, *The problem of hidden variables in quantum mechanics*, J. Math. Mech. **17**, 59 (1967), doi:10.1512/iumj.1968.17.17004.
- [2] C. Budroni, A. Cabello, O. Gühne, M. Kleinmann, and J.-Å. Larsson, *Kochen-Specker contextuality*, Rev. Mod. Phys. **94**, 045007 (2022), doi:10.1103/RevModPhys.94.045007.
- [3] M. Howard, J. Wallman, V. Veitch, and J. Emerson, *Contextuality supplies the ‘magic’ for quantum computation*, Nature **510**, 351 (2014), doi:10.1038/nature13460.
- [4] R. Raussendorf, *Contextuality in measurement-based quantum computation*, Phys. Rev. A **88**, 022322 (2013), doi:10.1103/PhysRevA.88.022322.
- [5] A. A. Klyachko, *Coherent states, entanglement, and geometric invariant theory*, arXiv:quant-ph/0206012 (2002).
- [6] A. A. Klyachko, M. A. Can, S. Binicioğlu, and A. S. Shumovsky, *Simple test for hidden variables in spin-1 systems*, Phys. Rev. Lett. **101**, 020403 (2008), doi:10.1103/PhysRevLett.101.020403.
- [7] M. Araújo, M. T. Quintino, C. Budroni, M. Terra Cunha, and A. Cabello, *All noncontextuality inequalities for the n -cycle scenario*, Phys. Rev. A **88**, 022118 (2013), doi:10.1103/PhysRevA.88.022118.
- [8] S. Gupta, D. Saha, Z.-P. Xu, A. Cabello, and A. S. Majumdar, *Quantum contextuality provides communication complexity advantage*, Phys. Rev. Lett. **130**, 080802 (2023), doi:10.1103/PhysRevLett.130.080802.
- [9] M. Lostaglio, *Certifying quantum signatures in thermodynamics and metrology via contextuality of quantum linear response*, Phys. Rev. Lett. **125**, 230603 (2020), doi:10.1103/PhysRevLett.125.230603.
- [10] A. Cabello, S. Severini, and A. Winter, *Graph-theoretic approach to quantum correlations*, Phys. Rev. Lett. **112**, 040401 (2014), doi:10.1103/PhysRevLett.112.040401.
- [11] L. Lovász, *On the Shannon capacity of a graph*, IEEE Trans. Inf. Theory **25**, 1 (1979), doi:10.1109/TIT.1979.1055985.
- [12] E. Amselem, L. E. Danielsen, A. J. López-Tarrida, J. R. Portillo, M. Bourennane, and A. Cabello, *Experimental fully contextual correlations*, Phys. Rev. Lett. **108**, 200405 (2012), doi:10.1103/PhysRevLett.108.200405.
- [13] G. Cañas, E. Acuña, J. Cariñe, J. F. Barra, E. S. Gómez, G. B. Xavier, G. Lima, and A. Cabello, *Experimental demonstration of the connection between quantum contextuality and graph theory*, Phys. Rev. A **94**, 012337 (2016), doi:10.1103/PhysRevA.94.012337.
- [14] D. E. Knuth, *The sandwich theorem*, Electron. J. Comb. **1**, A1 (1994), doi:10.37236/1193.
- [15] B. D. McKay and A. Piperno, *Practical graph isomorphism, II*, J. Symb. Comput. **60**, 94 (2014), doi:10.1016/j.jsc.2013.09.003; graph databases at <https://users.cecs.anu.edu.au/~bdm/data/graphs.html>.
- [16] A. A. Hagberg, D. A. Schult, and P. J. Swart, *Exploring network structure, dynamics, and function using NetworkX*, in Proc. 7th Python in Science Conf. (SciPy2008), pp. 11–15 (2008), doi:10.25080/TCWV9851.
- [17] S. Diamond and S. Boyd, *CVXPY: A Python-embedded modeling language for convex optimization*, J. Mach. Learn. Res. **17**, 1 (2016); <https://www.jmlr.org/papers/v17/15-408.html>.
- [18] B. O’Donoghue, E. Chu, N. Parikh, and S. Boyd, *Conic optimization via operator splitting and homogeneous self-dual embedding*, J. Optim. Theory Appl. **169**, 1042 (2016), doi:10.1007/s10957-016-0892-3.
- [19] D. H. Bailey and H. R. P. Ferguson, *Numerical results on relations between numerical constants using a new algorithm*, Math. Comput. **53**, 649 (1989), doi:10.1090/S0025-5718-1989-0979934-9; see also D. H. Bailey and D. J. Broadhurst, *Parallel integer relation detection*, Math. Comput. **70**, 1719 (2001), doi:10.1090/S0025-5718-00-01278-3.
- [20] M. Nakata, *A numerical evaluation of highly accurate multiple-precision arithmetic version of semidefinite programming solver: SDPA-GMP, -QD and -DD*, in Proc. 2010 IEEE Multi-Conf. Systems and Control, pp. 29–34 (2010), doi:10.1109/CACSD.2010.5612693.
- [21] G. Molina-Terriza, J. P. Torres, and L. Torner, *Twisted photons*, Nature Phys. **3**, 305 (2007), doi:10.1038/nphys607.
- [22] X.-M. Hu, J.-S. Chen, B.-H. Liu, Y. Guo, Y.-F. Huang, Z.-Q. Zhou, C.-F. Li, and G.-C. Guo, *Experimental test of compatibility-loophole-free contextuality with spatially separated entangled qutrits*, Phys. Rev. Lett. **117**, 170403 (2016), doi:10.1103/PhysRevLett.117.170403.
- [23] R. W. Spekkens, *Contextuality for preparations, transformations, and unsharp measurements*, Phys. Rev. A **71**, 052108 (2005), doi:10.1103/PhysRevA.71.052108.
- [24] D. M. Cvetković, M. Doob, and H. Sachs, *Spectra of Graphs: Theory and Applications* (Academic Press, New York, 1980).
- [25] G. Kirchmair, F. Zähringer, R. Gerritsma, M. Kleinmann, O. Gühne, A. Cabello, R. Blatt, and C. F. Roos, *State-independent experimental test of quantum contextuality*, Nature **460**, 494 (2009), doi:10.1038/nature08172.
- [26] M. Jerger, Y. Reshitnyk, M. Oppliger, A. Potočník, M. Mondal, A. Wallraff, K. Goodenough, S. Wehner, K. Juliusson, N. K. Langford, and A. Fedorov, *Contextuality without nonlocality in a superconducting quantum system*, Nat. Commun. **7**, 12930 (2016), doi:10.1038/ncomms12930.
- [27] R. Lapkiewicz, P. Li, C. Schaeff, N. K. Langford, S. Ramelow, M. Wieśniak, and A. Zeilinger, *Experimental non-classicality of an indivisible quantum system*, Nature **474**, 490 (2011), doi:10.1038/nature10119.
- [28] S. Dogra, K. Dorai, and Arvind, *Experimental demonstration of quantum contextuality on an NMR qutrit*, Phys. Lett. A **380**, 1397 (2016), doi:10.1016/j.physleta.2016.02.017.
- [29] E. Amselem, M. Rådmark, M. Bourennane, and A. Cabello, *State-independent quantum contextuality with single photons*, Phys. Rev. Lett. **103**, 160405 (2009), doi:10.1103/PhysRevLett.103.160405.
- [30] P. Gnaniński, M. Rosicka, R. Ramanathan, K. Horodecki, M. Horodecki, P. Horodecki, and S. Severini, *Linear game non-contextuality and Bell inequalities — a graph-theoretic approach*, New J. Phys. **18**, 045020 (2016), doi:10.1088/1367-2630/18/4/045020.
- [31] M. Ray, N. G. Boddu, K. Bharti, L.-C. Kwek, and A. Cabello, *Graph-theoretic approach to dimension witnessing*, New J. Phys. **23**, 033006 (2021), doi:10.1088/1367-

- 2630/abcacd.
- [32] Z.-P. Xu, X.-D. Yu, and M. Kleinmann, *State-independent quantum contextuality with projectors of nonunit rank*, New J. Phys. **23**, 043025 (2021), doi:10.1088/1367-2630/abef6e3.
 - [33] L. E. A. Porto, G. Ruffolo, R. Rabelo, M. Terra Cunha, and P. Kurzyński, *Trade-off relations between Bell non-locality and local Kochen-Specker contextuality in generalized Bell scenarios*, New J. Phys. **26**, 083028 (2024), doi:10.1088/1367-2630/ad7167.
 - [34] U. Tamer, Ö. E. Müstecaplıoğlu, A. Dizdar, and Z. Gedik, *Quad-C5 contextuality: code and data*, Zenodo (2026), doi:10.5281/zenodo.20465134.